Rishi Babu

Postdoctoral Research Associate • Michigan State University

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ORCID Website

Academic Positions

Postdoctoral Research Associate Feb 2024 – present
Michigan State University, Department of Physics & Astronomy East Lansing, MI
Supervisors: Mehr Un Nisa, Tyce DeYoung, Nathan Whitehorn, Kirsten Tollefson
Gamma-ray and neutrino studies of PeVatron candidates with the HAWC and IceCube Observatories

Guest Lecturer May 2025 – present
Michigan State University East Lansing, MI

Education

Ph.D., Physics Feb 2024
Michigan Technological University Houghton, MI
Thesis: *Study of Particle Accelerators in the Universe with the HAWC Observatory*
Advisor: Petra Huentemeyer

M.S., Physics Apr 2021
Michigan Technological University Houghton, MI

B.Tech., Engineering Physics May 2017
National Institute of Technology Calicut Calicut, India

Honors & Awards

Co-PI, NASA Fermi Guest Investigator Program, Cycle 18 2025
Open Scholarship Fellowship, Michigan State University / NASA 2025
Henes Graduate Research Fellowship, Michigan Technological University 2024
Dean's Award for Outstanding Scholarship, Michigan Technological University 2024
Outstanding Graduate Teaching Award, Michigan Technological University 2024
American Physical Society Travel Grant 2022, 2023
Recognition for Exceptional Teaching by the Provost, Michigan Technological University 2021

Scientific Collaboration Leadership

Data Release Technical Lead, HAWC Collaboration 2026 – present
Plan and coordinate HAWC public data releases; liaison between the collaboration and external scientists.

Data & Algorithms Working Group Leader, HAWC Collaboration 2025 – present
Coordinate the work of 50+ scientists; oversee new reconstruction pipelines, data integrity and reproducibility; represent the working group on the HAWC Collaboration Board.

Galactic Science Working Group Co-Leader, HAWC Collaboration 2025 – 2026
Overee analysis review and preparation of results for publication.

Grants Experience

Co-PI, NASA Fermi Guest Investigator Program, Cycle 18 2025
Open Scholarship Fellowship, Michigan State University / NASA 2025
Proposed the HAWC data-release schema and tools for public data access.

Contributor, NSF 26-512: Unlocking Dataset Value for AI-Enabled Scientific Discovery (HAWC) 2026

Contributor, NSF Windows on the Universe: Multi-Messenger Astrophysics, MSU 2024, 2025

Press Coverage

Observation of the Galactic Center PeVatron Beyond 100 TeV with HAWC 2024
Corresponding author

- Featured in a Los Alamos National Laboratory press release.
- Covered by CERN Courier, Universe Today, and ScienceAlert.

Professional Service

Experimental Membership Roles

- Data and Algorithms Working Group Lead, HAWC Observatory 2025-
- Public Data Release Lead, HAWC Observatory 2026-
- Galactic Working Group Co-Leader, HAWC Observatory 2025
- Member, IceCube Observatory 2024-
- Member, HAWC Observatory 2020-

Meeting and Conference Organization

- APS Global Physics Summit 2025, Session Chair 2025
- APS 2023, Session Chair 2023
- Fermi-Veritas-HAWC Symposium, Co-organizer 2019

Reviewer Service

- NASA Fermi Guest Investigator Program (Cycle 19) 2025
- NASA Citizen Science Seed Funding Program (CSSRP) 2025
- Referee, *Simulation Modelling Practice and Theory* (Elsevier) 2026
- Internal Reviewer for HAWC and IceCube collaborations 2023-

Departmental Service

- Michigan Technological University, Graduate Student Government 2020-2021
- National Institute of Technology, Students Affairs Council 2014-2017

Invited Talks, Colloquia & Lectures

[1] "Multi-messenger Astrophysics with Neutrinos," invited lecture, Scientific and Technological Research Council of Türkiye (TÜBİTAK), Ankara, Turkey, 2026.

[2] "Understanding the Enigmatic Particle Accelerators in the Universe with Multi-messenger Observations," invited talk, Wright State University, OH, 2025.

[3] "Understanding the Enigmatic Particle Accelerators in the Universe with Multi-messenger Observations," colloquium, Michigan State University, MI, 2025.

Contributed Talks & Posters

[1] "On the TeV Emission from the Neutron Star Low-Mass X-ray Binary GX 13+1," Compact Objects in Michigan and Ontario, MSU, 2026.

[2] "Multi-wavelength Analysis of Unassociated 1LHAASO Sources using the HAWC Observatory," Lepton-Photon 2025, Madison, WI.

[3] "Improving Gamma-ray Source Searches with Image Processing," ICRC 2025, Geneva, Switzerland (poster, virtual).

[4] "Correlation of Neutrino Emission with Ultra-High-Energy Gamma Rays from LHAASO Dark Sources using IceCube and HAWC," APS March Meeting 2025, CA.

[5] "Analysis Results of HAWC Data Corresponding to 1LHAASO Sources Exclusively Linked to Fermi-LAT-Detected GeV Gamma Rays," TeVPA 2024, Chicago, IL.

[6] "HAWC Study of the Extended TeV Emission from HESS J1813-178," 11th International Fermi Symposium, 2024, MD.

[7] "High-Energy Gamma-Ray Emission from HESS J1809-193: Morphological and Spectral Studies with HAWC," ICRC 2023, Nagoya, Japan.

[8] "HAWC Analysis of HESS J1809-193: A Potential Hadronic PeVatron?," APS April Meeting 2023, Minneapolis, MN.

[9] "HAWC Study of HESS J1809-193: A PeVatron Candidate in a Source-Rich Region," APS April Meeting 2022, New York, NY.

Teaching Experience

Part-Time Lecturer May 2025 – present
Michigan State University East Lansing, MI

- Substitute instructor, PHY 294 Honors Electricity and Magnetism.

Graduate Teaching Assistant 2020 – 2024
Michigan Technological University Houghton, MI

- Instructor, PH 1600 Introductory Astronomy Laboratory (three years).
- Laboratory instructor, PH 1361 and PH 2261 Introductory Experimental Physics (honors sections).
- Teaching assistant and substitute lecturer, PH 4620 Galactic Astrophysics and PH 4630 Particle Astrophysics.

Recognized with the MTU Outstanding Graduate Teaching Award (2024) and the Provost's Recognition for Exceptional Teaching (2021).

Research Experience

HAWC Collaboration 2020 – present

- Co-led the first observation of emission above 100 TeV from the Galactic Center region near Sgr A* [J4].
- Led the spectral and morphological study of HESS J1809−193 as a candidate hadronic PeVatron [J5], and the study of the pulsar wind nebula G32.64+0.53 as a PeVatron candidate using a time-dependent diffusion model [J2].
- Developed a fast source-detection method using Fast Fourier Transforms and image processing, about 300× faster than standard methods [J3].
- Developed diffuse gamma-ray emission templates used in collaboration analyses; developer of the `3ML` and `astromodels` analysis packages.

IceCube Collaboration 2024 – present

- Co-led the first joint HAWC–IceCube search for neutrino emission from the Galactic Center [J1].
- Multi-messenger searches for neutrinos from unidentified gamma-ray sources and from star-forming galaxies.
- Developing a GPU-based, physics-informed neural network (JAX/Equinox) for muon-neutrino track reconstruction.
- Searching for tau-neutrino signatures in 12 years of IceCube data with graph attention transformers.

Research Mentoring

Graduate students

- Akshay Zindal (MSU) 2026 – present
- Hitesh Malhotra (MSU) 2026 – present
- Manish Khanal (University of Utah) 2025 – 2026

- Brandon Henke (MSU) 2025 – present
- Joselyn Winda (MSU) 2025 – present
- Alejandra Granados (MSU) 2024 – present
- Ian Herzog (MSU) 2024 – 2026
- Youngwan Son (SKKU, South Korea); now postdoctoral researcher, SKKU 2022 – 2025

Undergraduate and post-baccalaureate students

- Ved Vinod, post-baccalaureate (MSU) 2026 – present
- Palmer Wentworth, undergraduate (MSU) 2024 – present

Publications

Full list: Google Scholar. Collaboration papers list the collaboration as author; my role is given in parentheses.

Lead or corresponding author

[J1] HAWC & IceCube Collaborations (lead author), "Resolving the Galactic Center Region's Gamma-Ray Emission and Searching for its Neutrino Counterpart with HAWC and IceCube," arXiv:2609.22577, submitted to *ApJL*.

[J2] HAWC Collaboration (lead author), "HAWC Study on the Ultra-High-Energy Gamma-Ray Emissions from the Pulsar Wind Nebula G32.64+0.53," *ApJ* **1009**, 47 (2026).

[J3] **R. Babu**, "Improving Gamma-ray Source Searches with Fast Fourier Transforms and Image Processing," *PoS* (ICRC2025) 557 (2025).

[J4] HAWC Collaboration (corresponding author), "Observation of the Galactic Center PeVatron Beyond 100 TeV with HAWC," *ApJL* **973**, L34 (2024).
Press: Los Alamos National Laboratory, CERN Courier, Universe Today, ScienceAlert.

[J5] HAWC Collaboration (lead author), "TeV Analysis of a Source Rich Region with HAWC Observatory: Is HESS J1809-193 a Potential Hadronic PeVatron?," *ApJ* **972**, 21 (2024).

Contributing author (selected)

[C1] R. Alfaro et al. (HAWC), "Investigating IceCube Neutrino Alerts with the HAWC γ-Ray Observatory," accepted, *ApJ*.

[C2] R. Alfaro et al. (HAWC), "Analysis of the Emission and Morphology of the Pulsar Wind Nebula Candidate HAWC J2031+415," *ApJ* **975** (2024).

[C3] R. Alfaro et al. (HAWC), "Testing the Molecular Cloud Paradigm for Ultra-High-Energy Gamma-Ray Emission from the Direction of SNR G106.3+2.7," *A&A* **691** (2024).

[C4] R. Alfaro et al. (HAWC), "Ultra-high-energy gamma-ray bubble around microquasar V4641 Sgr," *Nature* **634**, 557–560 (2024).

In preparation (lead or corresponding author, under collaboration review)

[P1] HAWC, "Evidence of a TeV Gamma-Ray Bubble from GX 13+1 near HESS J1813-178 with the HAWC Observatory," to be submitted to *ApJ*.

[P2] HAWC, "Astrophysical Source Search using Image Processing: Application to HAWC Observatory," to be submitted to *ApJ*.

[P3] HAWC, "HAWC Detection of 1LHAASO J2027+3657 and its Connection to the Star-Forming Region S106."

[P4] IceCube, "Search for Neutrino Tracks from Starburst Galaxies using 12 Years of IceCube Data."

[P5] IceCube & HAWC, "Search for Neutrino Tracks and γ-Rays from LHAASO Unidentified Sources using IceCube and HAWC."

Outreach

Guest Speaker, Skype a Scientist	2026
Presenter, IceCube Masterclass, Michigan State University	2024 – present
Presenter, Astronomical Horizons Public Lecture Series, Abrams Planetarium (MSU)	2025 – present
Presenter, MSU Science Festival	2025 – present

Computational Skills

Languages	Python, C++, SQL, R, MATLAB
ML & GPU	JAX/Equinox, PyTorch, TensorFlow, CUDA; graph neural networks, transformers, physics-informed neural networks
Astro & statistics	3ML, astromodels, HEALPix, maximum-likelihood and Bayesian inference, FFT-based image analysis
Computing	HPC and distributed computing (HTCondor, Slurm), Linux, Git

References

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